5 Backups and your data

FULL

12. Your database (MariaDB, which speaks MySQL)

Quick version

No time to read it all? This is the short version. The full guide is below.

- Your garden's database is called `juns_garden_db`. It runs in the container `garden-db` and is never reachable from the network.
- Your plants are in the table `data_source`.
- Look inside in your browser with Adminer: menu `M`, then `8`, then open http://localhost:8082. Stop it afterwards with `bash garden.sh dbweb-stop`.
- Or type SQL: `bash garden.sh db`.
- Back up first (menu `6`) every time you change data by hand. Made a mistake? Restore that backup (menu `7`).
- Always use `WHERE` on `UPDATE` and `DELETE`. Without it, every row changes.
- Most changes are easier and safer in the plant admin (`/streamlit/`).

What it is

A database is a set of tables, like spreadsheets that programs can search fast. Your garden's database is called `juns_garden_db`. It runs in the container `garden-db` and is never reachable from the network: only the garden's own programs talk to it.

The main tables

Table	What it holds
`data_source`	Your plants: the catalogue (one row per plant)
`future_plants`	The wish list and plant ideas
`plant_barcodes`	Barcodes linked to plants (the app's barcode lookup)
`flower_bed_qr`	The flower beds and their QR codes
`garden_blog`	Blog posts
`weather`, `weather_forecast`	Weather readings and forecasts
`user_favorites`, `user_likes`, `user_done`, `user_notes`, `user_prefs`	Visitors' likes, favourites and notes
`plant_purchase`, `plant_quantity_log`	What was bought, and how many
`wiki_draft`	Drafts waiting in the wiki queue
`vocab_alias`, `vocab_proposal`	The word lists the tag reader uses
`garden_users`, `garden_sessions`, `login_attempts`	Logins. Passwords are stored only as Argon2 hashes.

The wiki keeps its pages in a second database, `junsgarden_wiki`.

Look inside: two ways

The easy way, in your browser (Adminer): menu `M`, then `8`. Open http://localhost:8082 and fill in: System **MySQL**, Server **db**, Username **dbadmin**, Database **juns_garden_db**. For the password, see Settings, or type `grep ^DB_PASSWORD= .env` in the garden folder. Stop it afterwards: `bash garden.sh dbweb-stop`.

The command line (SQL): `bash garden.sh db`, then try:

```
SHOW TABLES;
SELECT COUNT(*) FROM data_source;
SELECT id, common_name FROM data_source ORDER BY id DESC LIMIT 10;
exit
```

(If a column name is different, `DESCRIBE data_source;` lists the real ones.)

Safe habits

1. **Back up first** (menu `6`), every time you change data by hand.
2. **Look before you change:** run the `SELECT` with the same `WHERE` first, and check it finds only the rows you mean.
3. **Always use `WHERE`** on `UPDATE` and `DELETE`. Without it, every row changes.
4. Most changes are easier and safer through the website's plant admin (`/streamlit/`) than by hand.
5. Made a mistake? Restore the backup you just made (menu `7`).

Changing the shape of a table (advanced)

Adding a column: back up, `ALTER TABLE ... ADD COLUMN ...`, add the same line to `db/schema.sql` so new installs have it, then change the Rust code that reads it (guide 11). One change at a time.

Read more (official)

- MariaDB SQL basics: https://mariadb.com/kb/en/basic-sql-statements/
- SELECT: https://mariadb.com/kb/en/select/
- Adminer: https://www.adminer.org

Jun's Garden · START-HERE/12-your-database.md

loved.
endured.
never forgotten.